Computer architecture and Organization

I/O Organization

M. Khademul Islam, PhD
Professor, Dept. of CSE, Rajshahi University, Bangladesh
Email: khademul.cse@ru.ac.bd, tel: +88-01727-786600
web: www.ru.ac.bd/cse/~mki

I/O Organization

Peripheral devices

- The input or output devices attached to the computer are also called peripherals.
- Among the most common peripherals are keyboards, display units and printers
- The peripherals that provide auxiliary storage for the system are magnetic disks and tapes.
- The devices that are under the direct control of computer are said to be connected online.

I/O Organization

I/O Interface

- Provides the method for transferring information between internal computing unit and external I/O devices.

Interface

- I/O interface requires the following things:
  - Special hardware components between the CPU and peripherals
  - Supervise and synchronize all input and output transfers

I/O Organization

Synchronous Data Transfer

- All data transfers occur simultaneously during the occurrence of a clock pulse
- Registers in the interface share a common clock with CPU registers

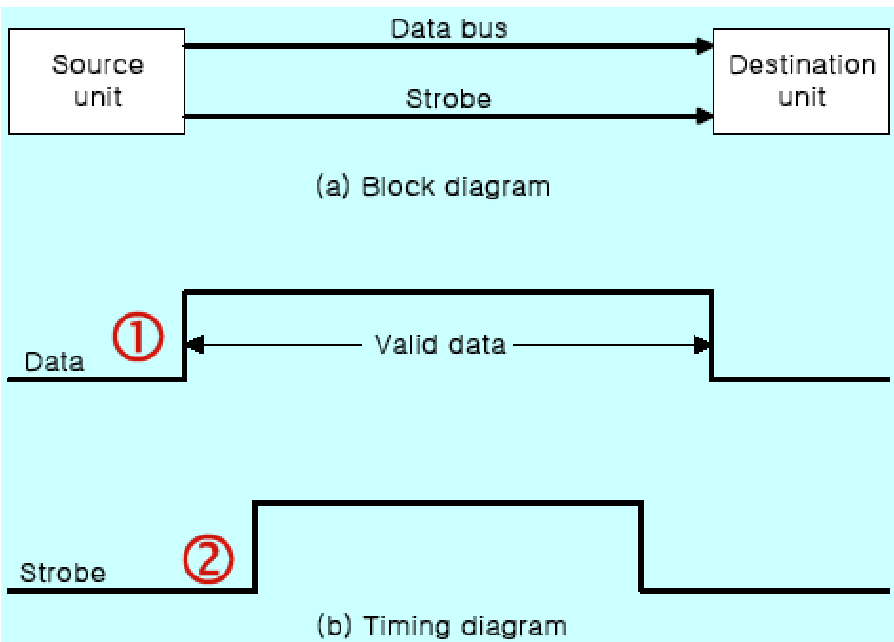
Asynchronous Data Transfer

- Internal timing in each unit (*CPU and Interface)* is independent
- Each unit uses its own private clock for internal registers

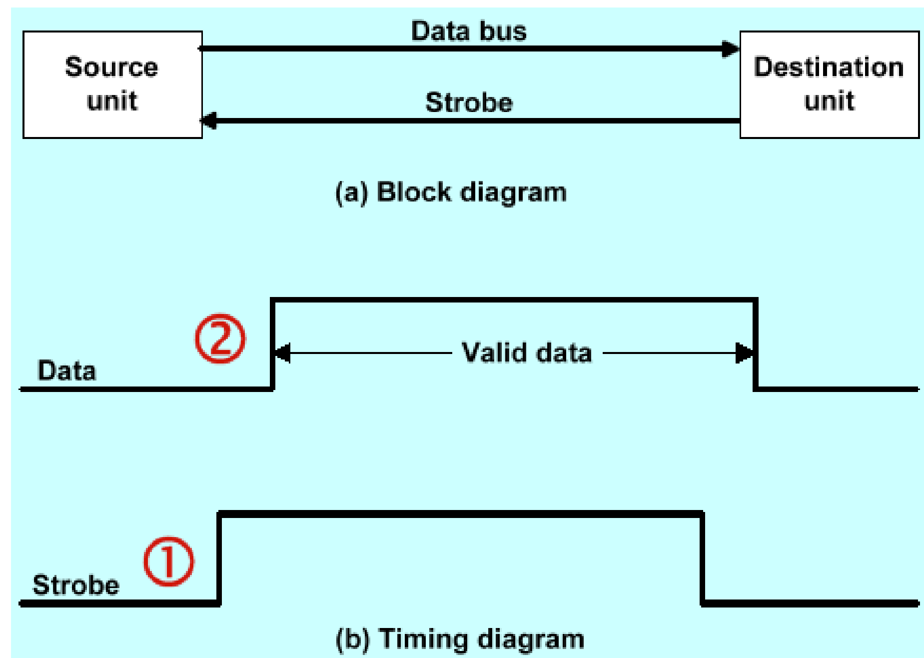
I/O Organization

Strobe : Control signal to indicate the time at which data is being transmitted

- Source-initiated strobe
- Destination-initiated strobe



Source-initiated strobe

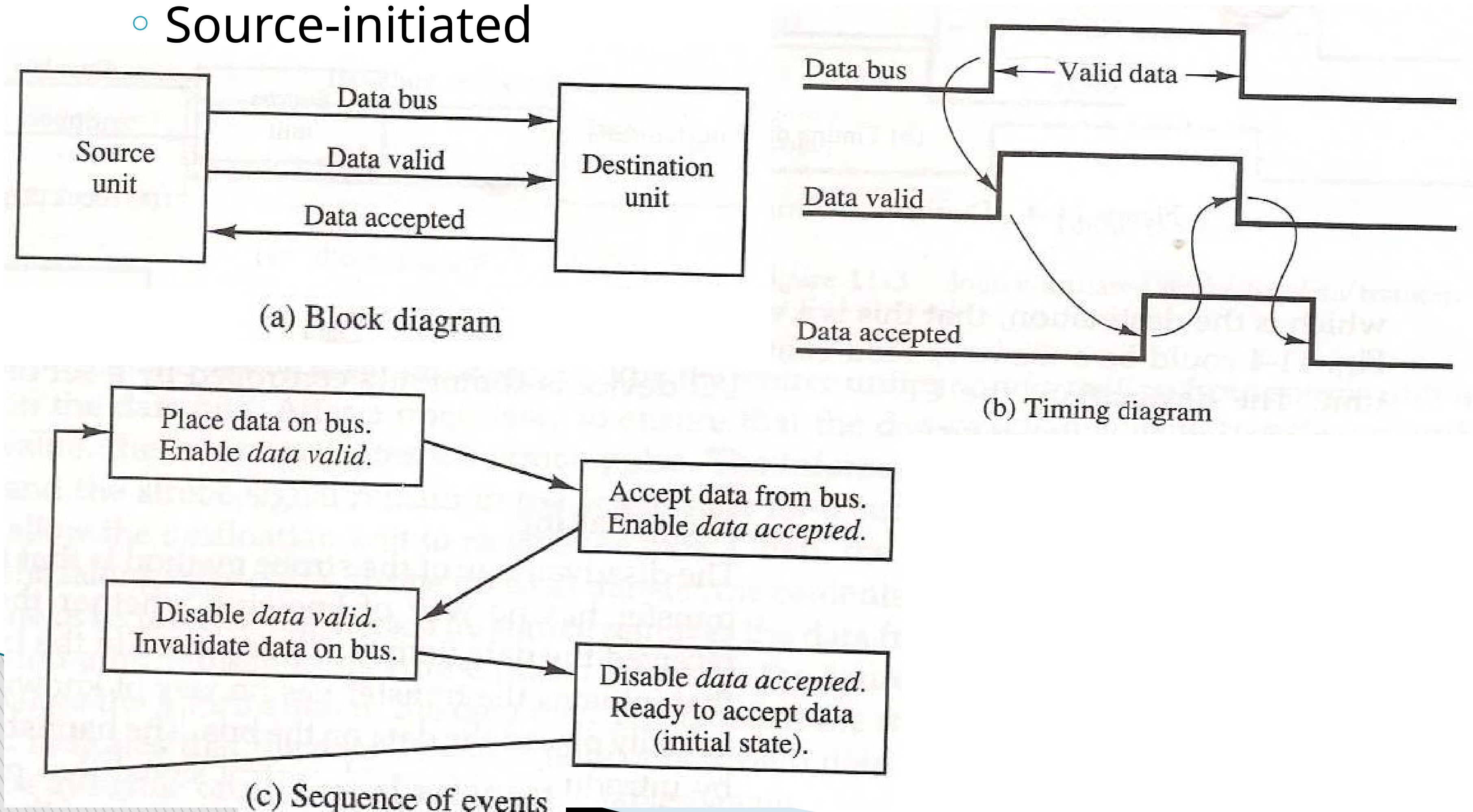


Destination-initiated strobe

I/O Organization

Handshake: Agreement between two units

- Source-initiated



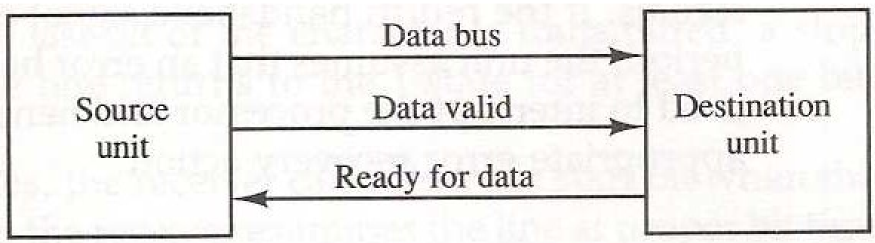
(a) Block diagram

(b) Timing diagram

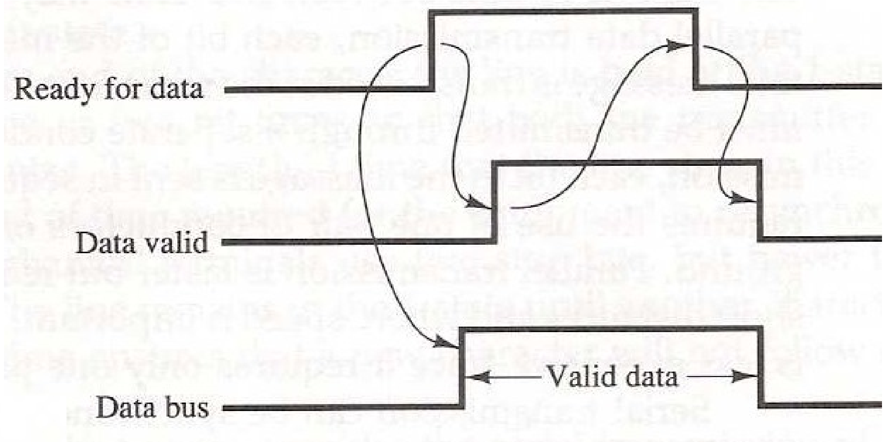
(c) Sequence of events

I/O Organization

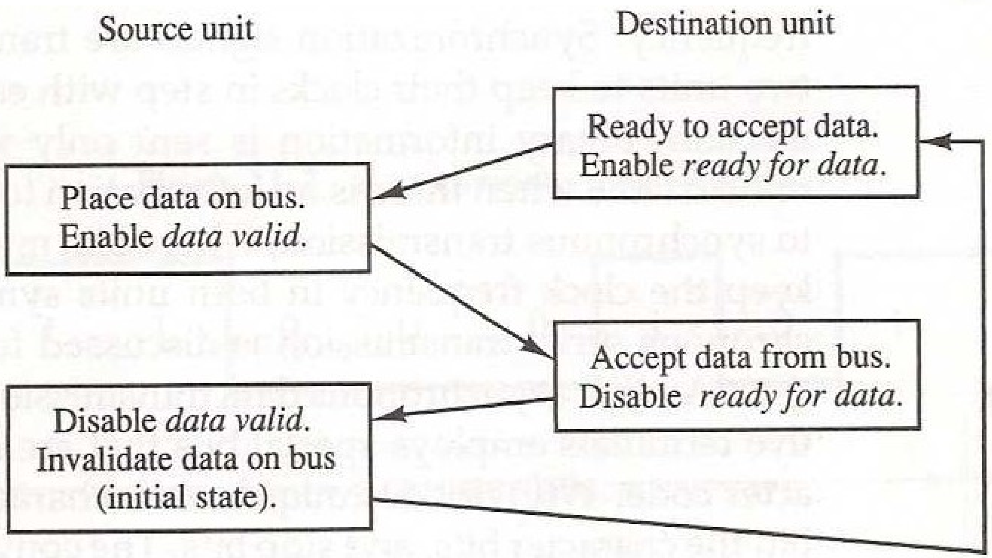
Destination initiated handshake



(a) Block diagram



(b) Timing diagram



(c) Sequence of events

I/O Organization

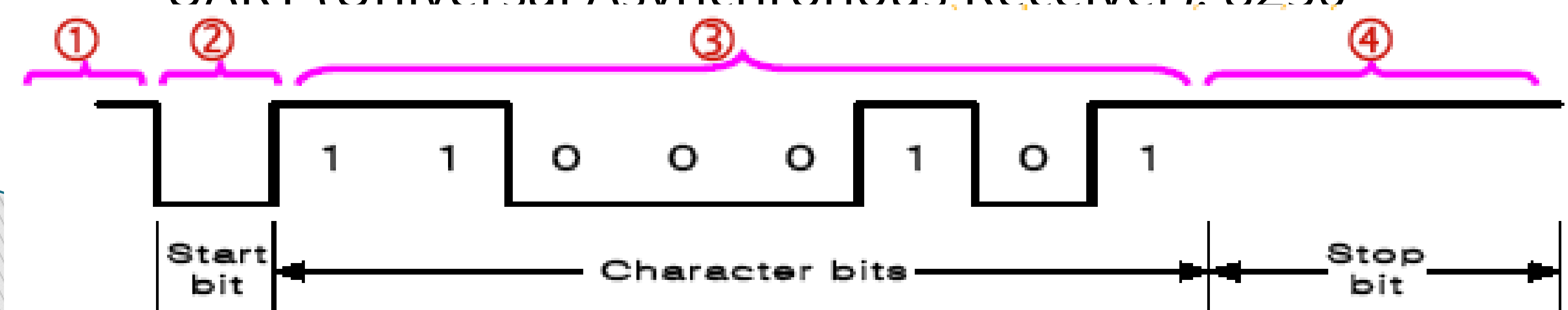
Asynchronous Data Transfer

- Special bits are inserted at both ends of the character code
- Each character consists of three parts :
- start bit : always “0”, indicate the beginning of a character
- character bits : data
- stop bit : always “1”

I/O Organization

Asynchronous transmission rules

- (1) When a character is not being sent, the line is kept in the 1-state
- (2) The initiation is detected from the start bit, which is always “0”
- (3) The character always follow the start bit
- (4) A stop bit is detected when the line returns to the 1-state for at least one bit time
- UART (Universal Asynchronous Receiver): 8250



I/O Organization

Interrupt-initiated I/O

- 1) Non-vectored : fixed branch address
- 2) Vectored : interrupt source supplies the branch address

Priority Interrupt

- Identify the source of the interrupts from simultaneous several sources
- Determine which one is to be serviced first from simultaneous multiple requests

I/O Organization

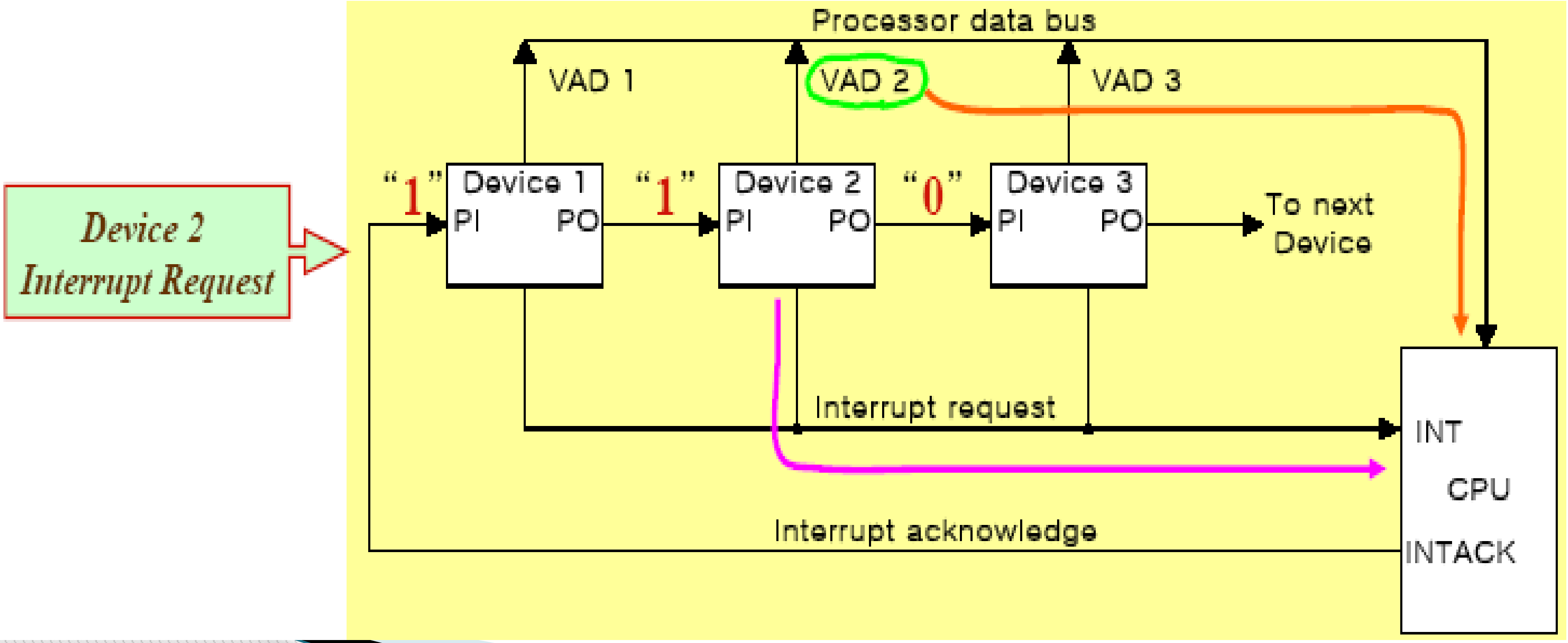
Software Polling

- Identify the highest-priority source by software means
- Polls the interrupt sources in sequence
- The highest-priority source is tested first

I/O Organization

Hardware priority interrupt

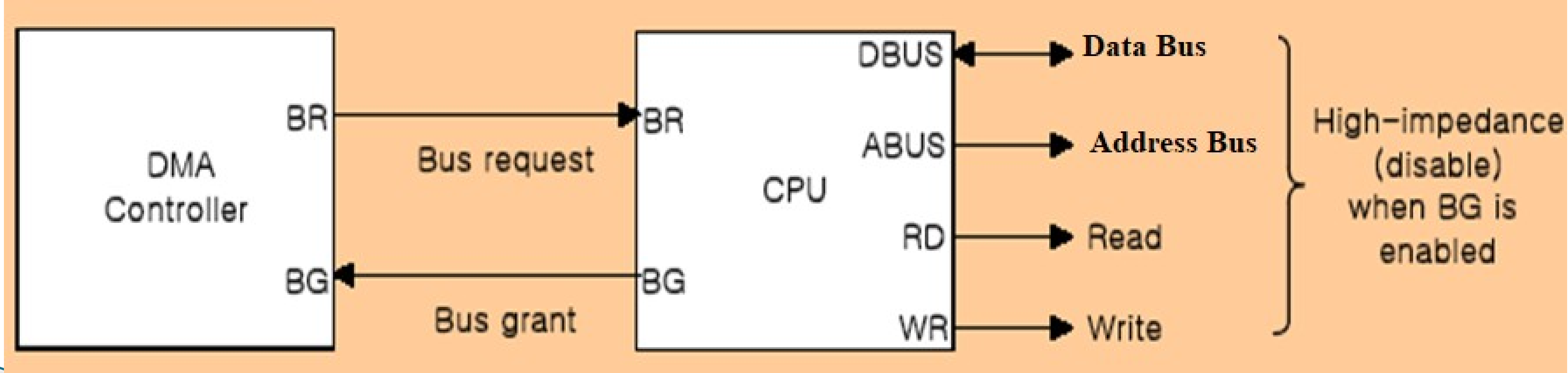
- Daisy-Chaining, parallel priority
  - Daisy -chaining



I/O Organization

Direct Memory Access (DMA)

- DMA controller takes over the buses to manage the transfer directly between the I/O device and memory
- Transfer Modes
  - Burst transfer : Block
  - Cycle stealing transfer : Word

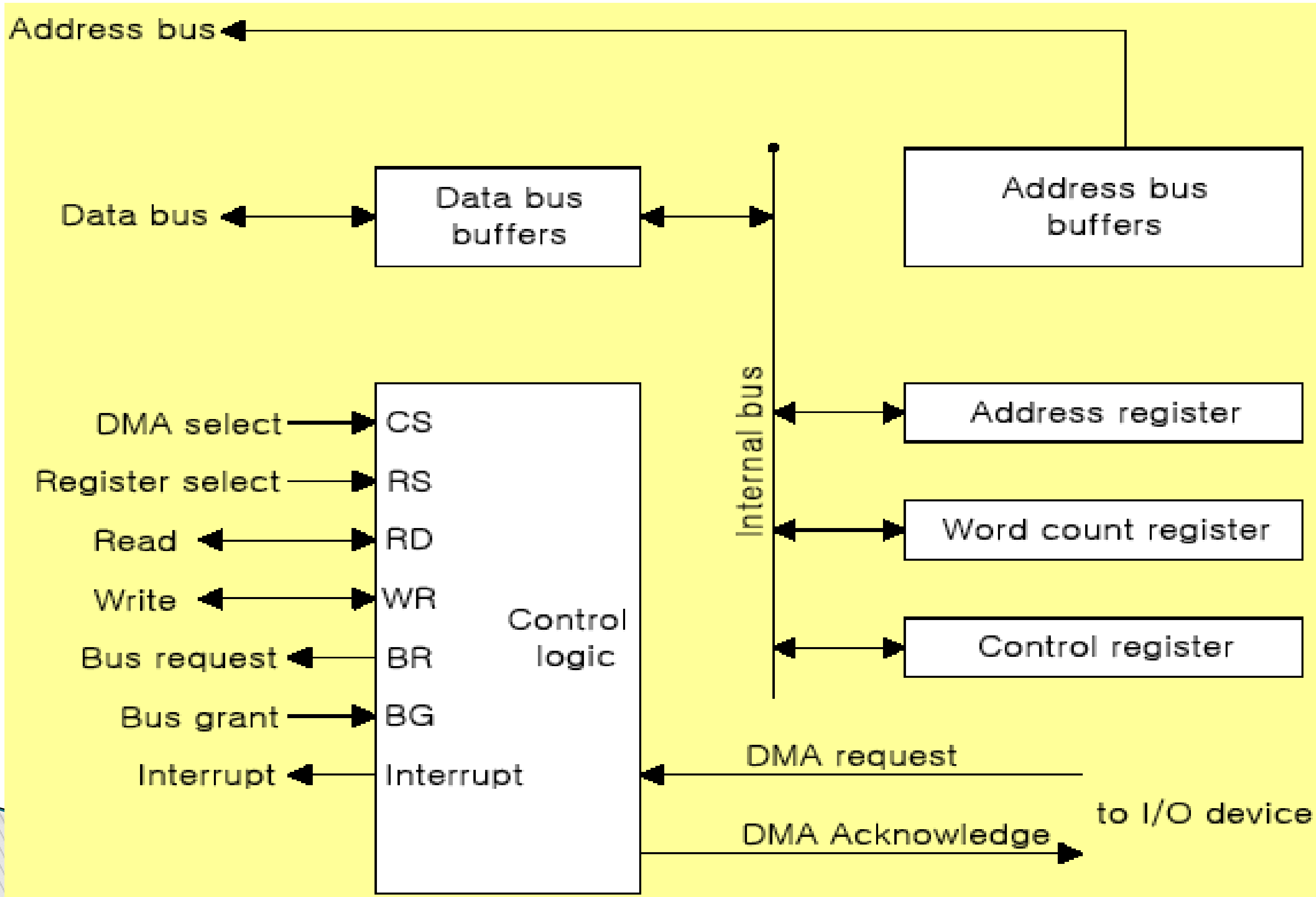


I/O Organization

DMA Controller (Intel 8237 DMAC) Initialization Process

- 1) Set Address register: memory address for read/write
- 2) Set Word count register: the number of words to transfer
- 3) Set transfer mode :
  - read/write,
  - burst/cycle stealing,
  - I/O to I/O,
  - I/O to Memory,
  - Memory to Memory
  - Memory search
  - I/O search
- 4) DMA transfer start : *next section*
- 5) EOT (End of Transfer): Interrupt

I/O Organization

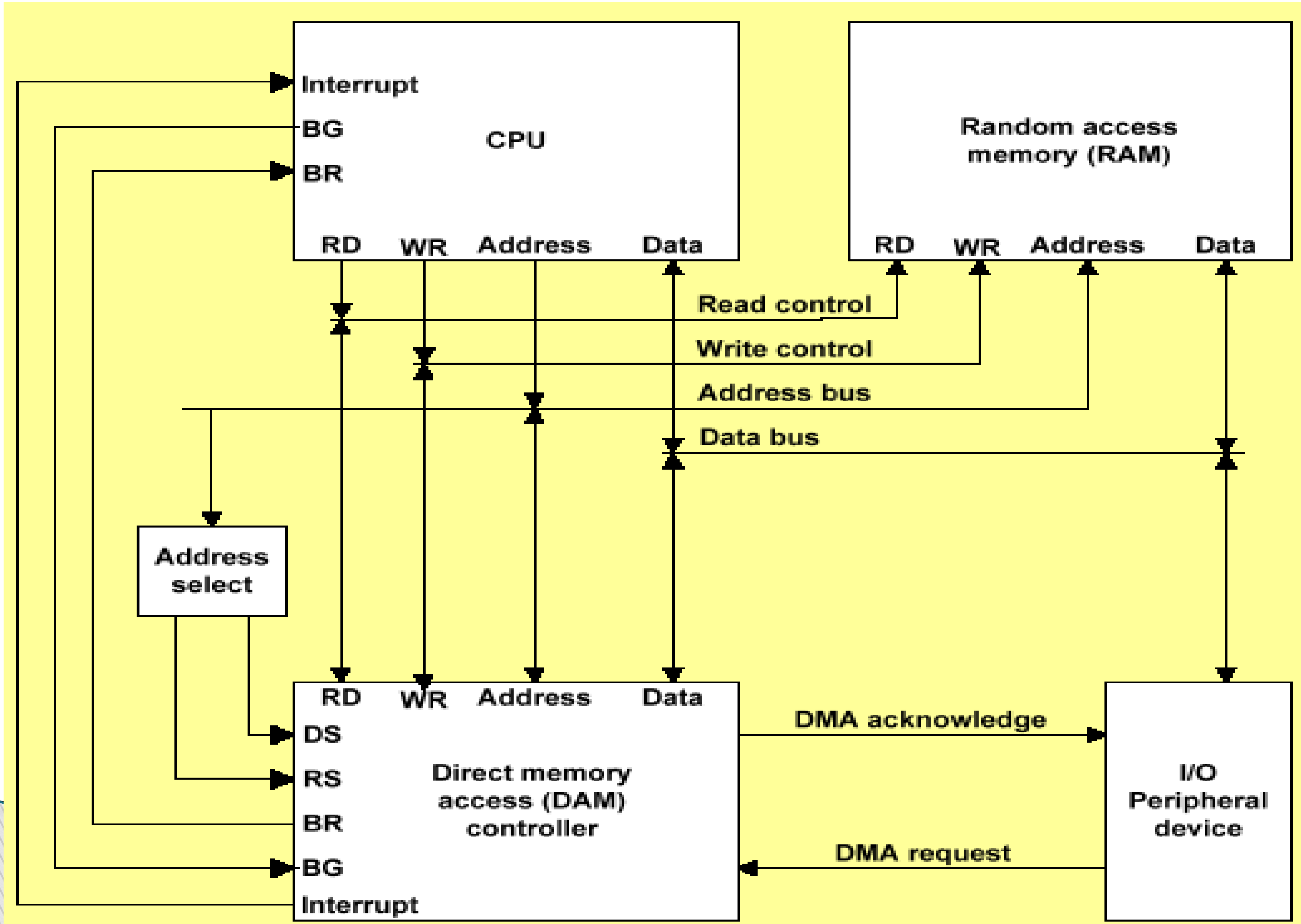


I/O Organization

DMA Transfer (I/O to Memory)

- 1) I/O Device sends a DMA request
- 2) DMAC activates the **BR line**
- 3) CPU responds with **BG line**
- 4) DMAC sends a DMA acknowledge to the I/O device
- 5) I/O device puts a word in data bus (*memory write)*
- 6) DMAC write a data to the address specified by **AR**
- 7) Decrement Word count register
- 8) Word count register = 0 (EOT interrupt)
- 9) Word count register ≠ 0 (DMAC checks the DMA request from I/O device)

I/O Organization



Bus Arbitration

Bus arbitration is required to decide which device gets control of the bus when multiple devices request it at the same time.

Why need?

- **To avoid conflicts**
  - If two devices drive the bus at the same time, data will be corrupted.
  - Arbitration ensures that only one master uses the bus at a time.
- **To maintain system stability**
  - If multiple masters try to access the bus simultaneously, the system may hang.
- **To ensure fair access**
  - All devices should get a fair chance to use the bus.
  - The device that requests first or has higher priority gets access.

Bus Arbitration

Why need?

- **To control priority**
  - Some devices (e.g., DMA controller) need higher priority.
- **To handle parallel requests**
  - When many devices request the bus at the same time, the arbitration algorithm decides who will get access.

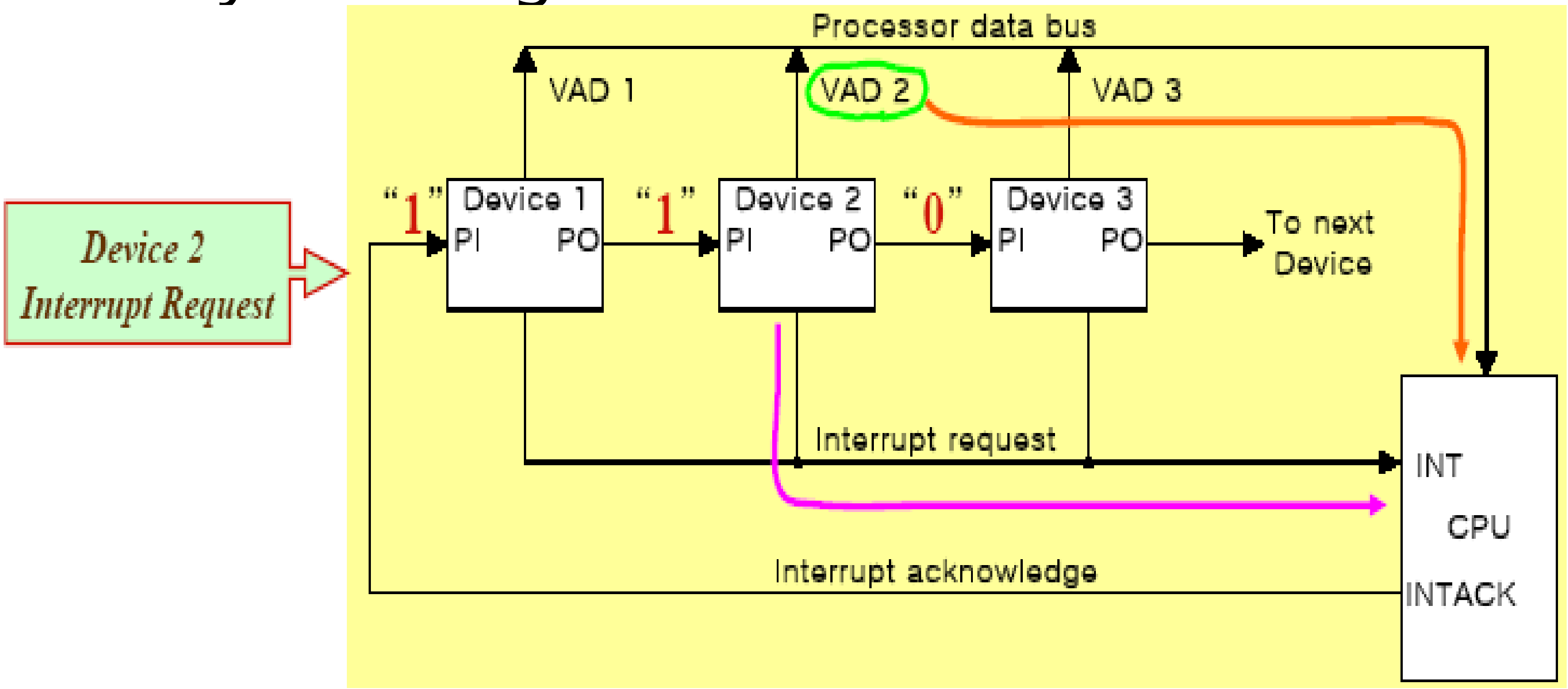
Bus Arbitration

Bus Arbitration Techniques

- Daisy Chaining
- Polling
- Independent Request

Bus Arbitration

- Daisy Chaining

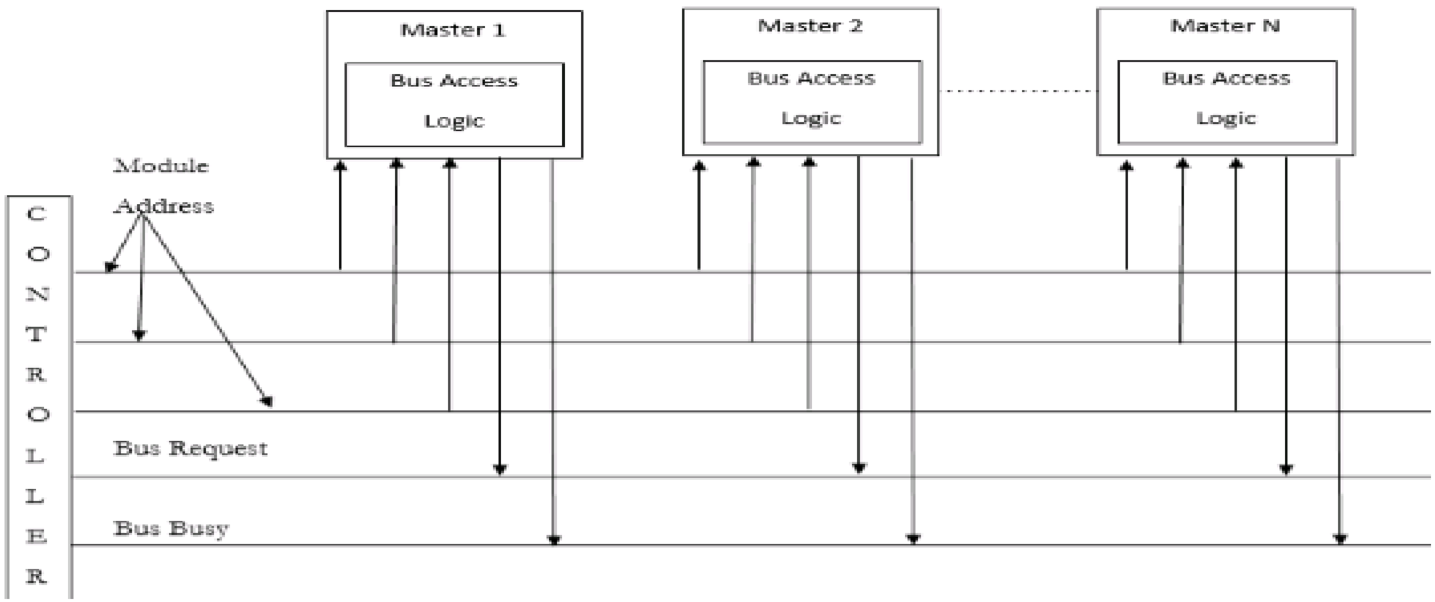


Bus Arbitration

- Daisy Chaining
- Advantages
  - Simple design
  - The user can add more devices anywhere along the chain
- Disadvantages
  - Priority is fixed
  - Lower-priority devices may face starvation
  - Delay increases in long chains
  - If one device fails then the entire system will stop working.

Bus Arbitration

- Polling



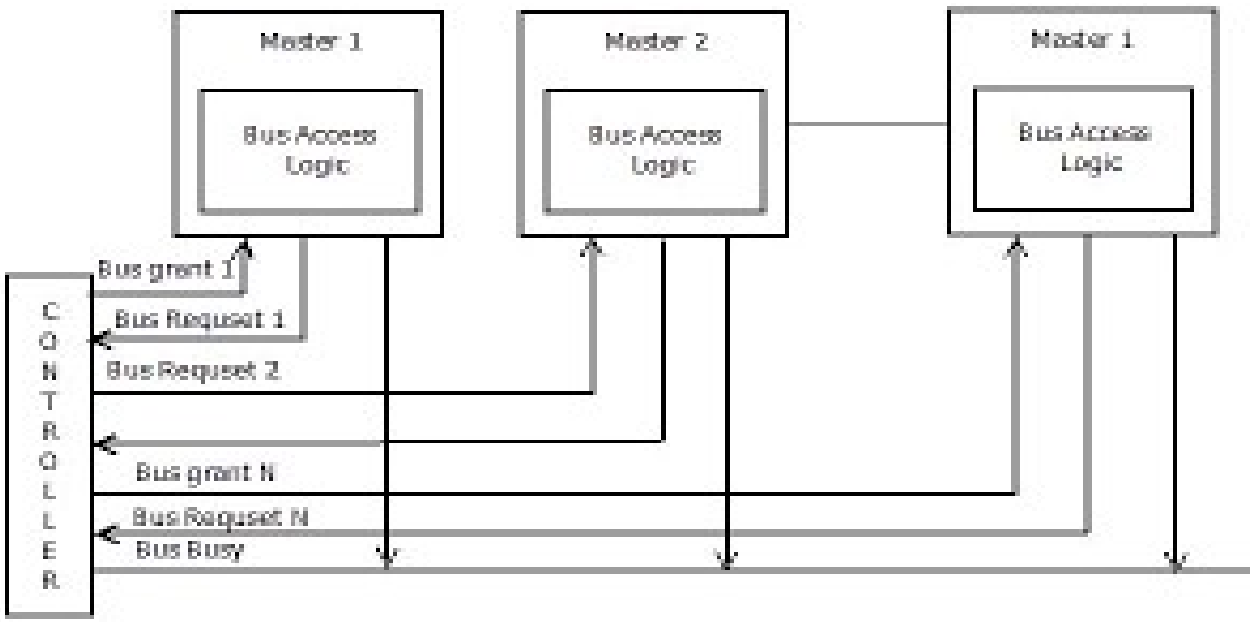
In this, the controller is used to generate the address for the master(unique priority), the number of address lines required depends on the number of masters

Bus Arbitration

- Polling
- Advantages
  - This method does not favor any particular device and processor.
  - The method is also quite simple.
  - If one device fails then the entire system will not stop working.
  - To reduce the starvation problem.
- Disadvantages
  - Adding bus masters is difficult as increases the number of address lines of the circuit.

Bus Arbitration

- Independent Request



The built-in priority decoder within the controller selects the highest priority request and asserts the corresponding bus grant signal.

Bus Arbitration

- Independent Request
- Advantages
  - Priority control is easy
  - This method generates a fast response.
  - To reduce the starvation problem.

- Disadvantages
  - Hardware cost is high as a large no. of control lines is required.